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Course: Calculus Beta

Assignment: MyLab Math - Uge 2

Use l'Hôpital's rule to find the following limit.

$$\lim_{x \rightarrow \infty} 12x^2 e^{-x}$$

L'Hôpital's rule states that if $f(x) \rightarrow \pm \infty$ and $g(x) \rightarrow \pm \infty$ as $x \rightarrow a$, if f and g are differentiable on an open interval I containing a , and if $g'(x) \neq 0$ on I if $x \neq a$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$, assuming that the limit on the right side of this equation exists.

Begin by rewriting the limit with only positive exponents. Thus, $\lim_{x \rightarrow \infty} 12x^2 e^{-x} = \lim_{x \rightarrow \infty} \frac{12x^2}{e^x}$. Let $f(x) = 12x^2$ and $g(x) = e^x$. Because $\lim_{x \rightarrow \infty} \frac{12x^2}{e^x}$ is an indeterminate form of type $\frac{\infty}{\infty}$, l'Hôpital's rule can be applied.

In order to apply l'Hôpital's rule, first find the derivative of the numerator, $f(x)$.

$$f'(x) = 24x$$

Next, find the derivative of the denominator, $g(x)$.

$$g'(x) = e^x$$

Therefore, by l'Hôpital's rule, $\lim_{x \rightarrow \infty} \frac{12x^2}{e^x} = \lim_{x \rightarrow \infty} \frac{24x}{e^x}$.

Notice that the limit still has the form $\frac{\infty}{\infty}$. Apply l'Hôpital's rule again. This time let $f(x) = 24x$ and $g(x) = e^x$. Find the derivative of the numerator, $f(x)$.

$$f'(x) = 24$$

Next, find the derivative of the denominator, $g(x)$.

$$g'(x) = e^x$$

So by l'Hôpital's rule, $\lim_{x \rightarrow \infty} \frac{12x^2}{e^x} = \lim_{x \rightarrow \infty} \frac{24x}{e^x} = \lim_{x \rightarrow \infty} \frac{24}{e^x}$.

Now find the limit.

$$\lim_{x \rightarrow \infty} \frac{24}{e^x} = 0$$

Therefore, $\lim_{x \rightarrow \infty} 12x^2 e^{-x} = 0$.